

Constant-coefficient abelian 1518-extensions of the $\mathbb{Z}/3$ shift are direct products

hence none refutes $1518 \Rightarrow 47, 614, 817, 3862$, and the ETP’s 15-element countermodel
is an extension of the same base with base-dependent coefficients

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1 Setting

Laws are numbered as in the Equational Theories Project (ETP), `teorth/equational_theories` at commit `88088faa`. Law 1518 is

$$x = (y \diamond y) \diamond (x \diamond (y \diamond x)),$$

and the four one-variable targets are

$$\begin{array}{ll} \text{E47 : } x = x \diamond (x \diamond (x \diamond x)), & \text{E614 : } x = x \diamond (x \diamond ((x \diamond x) \diamond x)), \\ \text{E817 : } x = x \diamond ((x \diamond x) \diamond (x \diamond x)), & \text{E3862 : } x \diamond x = (x \diamond (x \diamond x)) \diamond x. \end{array}$$

The smallest finite countermodel to $1518 \Rightarrow 47$ has 15 elements (Brox and Le Floch, Lean Zulip, October 2025), and the ETP’s 15-element table `Refutation939` refutes all four at once. Every 1518-magma with at most 7 elements satisfies all four (checked here on the complete census of Section 7).

Following the blueprint chapter “Magma cohomology”, an *extension* of a magma G by an abelian group M with $\alpha, \beta \in \text{End}(M)$ and $f : G \times G \rightarrow M$ is the magma on $G \times M$ with

$$(x, s) \diamond (y, t) = (x \diamond y, \alpha s + \beta t + f(x, y)). \tag{1}$$

The coefficients α, β are constant. If G and the linear magma $(M, \alpha s + \beta t)$ both satisfy a law E , the extension satisfies E if and only if f is an *E-cocycle*, a linear condition on f . Refuting $E \Rightarrow E'$ by this method means exhibiting an *E-cocycle* that is not an *E'-cocycle*.

In November 2024 this method produced the 12-element model for $879 \not\Rightarrow 4065$ (Tao, Lean Zulip, *Austin pairs*, message 484839156: base $(b, i) \diamond (c, j) = (b + c, j + 1)$ on $\mathbb{Z}/2 \times \mathbb{Z}/3$, fibre $\mathbb{Z}/2$, $\alpha = \beta = 1$) and failed for 1518 (Bolan 485027669; Tao 485133886: “perhaps sometimes the linear algebra just doesn’t work out”). Tao also observed (485146097) that since the targets have one variable, the base may be taken one-generated, and conjectured (485148288) that the only nontrivial one-generated 1518-magma satisfying the four targets is the cyclic shift $x \diamond y = y + 1$ on $\mathbb{Z}/3$.

*Computation and drafting with AI assistance under the author’s direction. Everything below is reproducible from a small standard-library script set; see Section 10.

2 Statement

Theorem 1. *Let G be the three-element magma $x \diamond y = y + 1$ on $\mathbb{Z}/3$, or the one-element magma. Let M be a finite abelian group and $\alpha, \beta \in \text{End}(M)$ such that $(M, \alpha s + \beta t)$ satisfies 1518. Then every 1518-cocycle $f: G \times G \rightarrow M$ is a 47-cocycle, a 614-cocycle, an 817-cocycle and a 3862-cocycle. Consequently no extension (1) of G refutes $1518 \Rightarrow E'$ for $E' \in \{47, 614, 817, 3862\}$.*

Corollary 2. *Let H be a finite 1518-magma and suppose an extension (1) of H by a finite M refutes $1518 \Rightarrow E'$ for one of the four targets. Then H contains a one-generated submagma with at least 9 elements. Dually, the same holds for $2054 \Rightarrow 255, 2847, 2644, 3456$ with the shift $x \diamond y = x + 1$.*

The corollary combines the theorem with Tao’s reduction and with a census (Section 7): up to isomorphism the one-generated 1518-magmas with at most 8 elements are the trivial magma and the $\mathbb{Z}/3$ shift. Section 6 removes the size bound: one-generated magmas satisfying 1518 and 3862 are trivial or the shift in any size, and the obstruction becomes unconditional for finite bases (Theorem 5).

3 Proof of Theorem 1

Step 1: the finite-fibre ring. Expanding 1518 on the linear magma $s \diamond t = \alpha s + \beta t$ and comparing the coefficients of x and y gives the identities in $\text{End}(M)$

$$\beta\alpha + \beta^3 = 1, \quad \alpha^2 + \alpha\beta + \beta^2\alpha = 0. \quad (2)$$

The first reads $\beta(\alpha + \beta^2) = 1$, so β is surjective; M is finite, hence β is bijective and $\alpha + \beta^2 = \beta^{-1}$, i.e. $\alpha = \beta^{-1} - \beta^2$. In particular α commutes with β . Substituting into the second identity and multiplying by β^2 yields

$$\beta^5 + \beta^3 - \beta^2 - 1 = (\beta^2 + 1)(\beta^3 - 1) = 0.$$

Conversely, these two facts imply (2). Finiteness of M is used only here, to pass from “surjective” to “bijective”; everything below holds verbatim for any abelian group M on which β is injective. Therefore M is a module over the commutative ring

$$R = \mathbb{Z}[b]/(b^5 + b^3 - b^2 - 1), \quad a := b^4 - b = b^{-1} - b^2, \quad b^{-1} = b^4 + b^2 - b, \quad (3)$$

via $b \mapsto \beta$, and then $a \mapsto \alpha$. R is free of rank 5 over $\mathbb{Z}$, and every R -module is an admissible fibre. Over an integral domain this recovers Nielsen’s classification (485154828): $\alpha = 0$, $\beta^3 = 1$, or $\beta = 1 - \alpha$, $\alpha^2 - 2\alpha + 2 = 0$.

Step 2: cocycle conditions over R . For any word w , the fibre component of w in the extension is $\sum_i P_{w,i}(a, b) s_i + \sum Q(a, b) f(\cdot, \cdot)$, the sum running over the non-variable subterms (blueprint). The linear magma over R satisfies 1518 and each of the four targets (the coefficient polynomials P of both sides agree in R ; checked), so the s_i terms cancel and the E -cocycle condition is a matrix D_E over R acting on the nine unknowns $f_{xy} = f(x, y)$. For every R -module M , $Z_E^2(G, M) = \ker(D_E: M^9 \rightarrow M^k)$.

Writing indices modulo 3, the 1518 condition at the base pair (x, y) is

$$(b^4 - b) f_{yy} + b^2 f_{yx} + b f_{x, x+1} + f_{y+1, x+2} = 0, \quad (4)$$

nine equations (when $x = y$ the first two terms merge into $(b^4 - b + b^2)f_{xx}$). The target conditions at x are

$$\text{E47: } b^2 f_{xx} + b f_{x,x+1} + f_{x,x+2} = 0, \quad (5)$$

$$\text{E614: } (b - b^4)f_{xx} + b f_{x,x+1} + f_{x,x+2} + b^2 f_{x+1,x} = 0, \quad (6)$$

$$\text{E817: } (1 + b^2 - b^3)f_{xx} + f_{x,x+2} + b f_{x+1,x+1} = 0, \quad (7)$$

$$\text{E3862: } b^3 f_{xx} + (b - b^4)f_{x,x+1} - f_{x+2,x} = 0. \quad (8)$$

Step 3: containment by certificate. For each target E' there is a 3×9 matrix $C_{E'}$ over R with $C_{E'} D_{1518} = D_{E'}$ entrywise in R . Then $D_{1518}f = 0$ implies $D_{E'}f = C_{E'} D_{1518}f = 0$ for every R -module M , which is the theorem. Writing r_{xy} for the 1518 row (4) at (x, y) , the 47 row (5) at $x = 0$ is

$$D_{47}(0) = r_{00} + (b - b^2 - b^4)r_{11} + (-1 + b + b^3)r_{20}. \quad (9)$$

The full certificates have 15 to 18 nonzero entries per target; they are listed in the accompanying file `12_certificate_3.json`, and the one-element base has 1×1 certificates in `12_certificate_1.json`.

Remark. The certificates were found, not guessed. Over the rank-5 ring the R -span of the nine 1518 rows is a $\mathbb{Z}$ -lattice of rank 35 in $\mathbb{Z}^{45}$, and membership of the target rows was decided by integer row reduction. Containment for all R -modules is in fact *equivalent* to this lattice containment (take $M = R^9 / \text{rowspan } D_{1518}$ and a finite quotient), so a failure would have produced an explicit finite countermodel. As a by-product, $Z_{1518}^2(G, M)$ has R -rank 2, and all of it consists of target cocycles.

4 The stronger statement: H^2 vanishes

Theorem 1 is a corollary of a cleaner fact. A *coboundary* is a function of the form $f(x, y) = g(x \diamond y) - \alpha g(x) - \beta g(y)$ for some $g: G \rightarrow M$; coboundaries are cocycles for every law, and an extension by a coboundary is isomorphic to the direct product via $(x, s) \mapsto (x, s + g(x))$ (blueprint, cohomology chapter). Let $E: M^3 \rightarrow M^9$ be the coboundary map $g \mapsto (g(y+1) - \alpha g(x) - \beta g(y))_{x,y}$, a 9×3 matrix over R .

Theorem 3. *For the $\mathbb{Z}/3$ shift G and every finite abelian group M with α, β as in Theorem 1, every 1518-cocycle $G \times G \rightarrow M$ is a coboundary: $H_{1518}^2(G, M) = 0$. Consequently every extension (1) of G by such a fibre that satisfies 1518 is isomorphic to the direct product $G \times (M, \alpha s + \beta t)$, and satisfies every law that both factors satisfy.*

Proof. There are matrices P (3×9 , two nonzero entries) and Q (9×9) over R with

$$EP + QD_{1518} = I_9 \quad \text{entrywise in } R. \quad (10)$$

If $D_{1518}f = 0$ then $f = EPf + QD_{1518}f = E(Pf)$, a coboundary, for every R -module M . The matrices are in `h2_certificate.json`; identity (10) is re-verified by multiplication in R and kernel-checked in `H2Cert.lean`. $\square$

Theorem 1 follows: the shift and every finite linear 1518-magma satisfy 47, 614, 817 and 3862, so their direct products do, so every extension does. The same argument gives the conclusion for *any* law satisfied by the shift and by all finite linear 1518-magmas.

Three checks and one contrast. At every root r of $b^5 + b^3 - b^2 - 1$ in $\mathbb{F}_p$, $p \leq 47$, one finds $\dim Z_{1518}^2 = \dim B^2 = 2$. Over the non-field fibre $\mathbb{Z}/4$ with $\beta = 1$, brute force gives 16 cocycles

and 16 coboundaries. The one-element base behaves differently: there H^2 can be nonzero (for $M = \mathbb{Z}/3$, $\beta = 1$, $\alpha = 0$ every constant is a cocycle and no nonzero constant is a coboundary), yet the containment of Theorem 1 still holds by its own certificate. Finally, for 879 over the affine base $3x + y + 1 \bmod 6$ with fibre $\mathbb{Z}/2$, $\alpha = \beta = 1$, one finds $\dim Z^2 = 7$, $\dim B^2 = 5$, so $\dim H^2 = 2$: that is the room the method used to produce the 12-element model, and it is absent for 1518 over the shift.

5 Consequences, and where the obstruction ends

The exact set of unreachable non-implications. By Theorem 3, a constant-coefficient 1518-extension of the shift satisfies every law E' that holds in the shift and in every finite linear 1518-magma; the second condition is decided once and for all in R (the coefficient polynomials of E' agree in R). Reading the row of 1518 in the ETP’s closed finite implication graph (`finite_graph.json`, fetched 2026-09-04): 1518 has 5 finite implications and 4689 finite non-implications, none unknown. Of the non-implications, 4347 are refuted by the shift itself, 329 by some finite linear 1518-magma on its own, and exactly

$$13 : \quad 47, 65, 375, 614, 632, 679, 817, 872, 879, 1491, 3862, 3915, 4118$$

are unreachable by every constant-coefficient extension of the shift: four one-variable laws (the targets) and nine two-variable laws. As a control, all 5 finite implications pass both checks. For the nine two-variable laws the statement is about this base only; the reduction to one-generated bases behind Corollary 2 needs a one-variable target.

The census, twice. Corollary 2 rests on the claim that the only one-generated 1518-magmas with at most 8 elements are the trivial one and the shift. Two independent tools agree: Mace4 (complete labelled enumeration at sizes 2–4, complete up to isomorphism at 5–7) and z3 with a direct reachability encoding of “generated by one element” (satisfiable only at size 3, where it returns the shift; unsatisfiable at sizes 2 and 4 to 8). The two tools also agree on the labelled counts 1, 6, 54, 1500 at sizes 2 to 5. Sizes 2 to 8 are moreover Lean theorems: the same encoding, discharged by `bv_decide`, whose trust rests on Lean’s verified LRAT checker run natively (Section 7).

The ETP witness is an extension of the shift with base-dependent coefficients. The 15-element table `Refutation939` has a unique nontrivial congruence, with three classes of five elements, and its quotient is isomorphic to the $\mathbb{Z}/3$ shift. Labelling each class by $\mathbb{Z}/5$ (the labelling is unique up to an affine relabelling of each fibre) writes the operation as

$$(x, s) \diamond (y, t) = (y + 1, \alpha_{x,y} s + \beta_{x,y} t + c_{x,y}),$$

with, for rows $x = 0, 1, 2$ and columns $y = 0, 1, 2$,

$$(\alpha_{x,y}, \beta_{x,y}, c_{x,y}) = \begin{pmatrix} (2, 4, 0) & (2, 3, 2) & (4, 1, 2) \\ (4, 2, 0) & (2, 4, 0) & (1, 4, 2) \\ (1, 3, 4) & (4, 2, 0) & (1, 2, 1) \end{pmatrix}.$$

The coefficients take the values $\alpha_{x,y} \in \{1, 2, 4\}$, $\beta_{x,y} \in \{1, 2, 3, 4\}$ and change with the base pair. This is exactly the general form of the blueprint’s Lemma “No counterexamples via linear extension” for 677, with the shift as base. By Theorem 3 no relabelling can make the coefficients constant. So the line is sharp: over the $\mathbb{Z}/3$ shift, constant coefficients give only direct products, while base-dependent coefficients give the ETP’s minimal countermodel. For 677 the obstruction survives base-dependent coefficients (Tao, June 2025); for 1518 over the shift it does not.

6 The one-generated bases are classified

The census left open what happens above size 8. It does not stay open. Write $S(x) = x \diamond x$.

Theorem 4. *In every magma satisfying 1518 and 3862, for every x and all u, v in the set $\{x, S(x), S(S(x))\}$ one has $u \diamond v = S(v)$, and $S(S(S(x))) = x$. Consequently the set is closed under $\diamond$, carries the multiplication table of the cyclic shift, has one element or three, and every one-generated magma satisfying 1518 and 3862 is the trivial magma or the $\mathbb{Z}/3$ shift.*

Vampire 5.1.0 found equational proofs of these nine equalities in milliseconds from 1518 and 3862 alone, by superposition and demodulation (9 to 32 proof lines each, `atp/eq*.proof`); they were transcribed into core Lean 4.33.1 (`lean/OneGenerated1518.lean`): the table and the closure of words in x are proved with no axioms at all, and the one-or-three-elements statement with `propext`, `Classical.choice`, `Quot.sound` only. The classification therefore needs neither finiteness nor the SAT census. This is Tao’s conjecture of 2024-11-29 in a stronger form: without finiteness and with one target instead of four. The target cannot be dropped: `Refutation939` is a one-generated 1518-magma with 15 elements, and it violates 3862 (60 of its 135 closure products leave the set). As a control, on all 41 319 six-element 1518-magmas the largest one-generated submagma has 3 elements.

Vampire also shows that under 1518 together with left injectivity and left surjectivity, which hold in every finite 1518-magma, the four targets 47, 614, 817, 3862 are pairwise equivalent (twelve theorems), and that equationally 3862 implies the other three. Hence:

Theorem 5 (full obstruction, finite bases). *Let H be a finite 1518-magma, M a finite abelian group with α, β making $\alpha s + \beta t$ a 1518-magma, and f a 1518-cocycle on H . If the extension $H \times_f M$ violates one of 47, 614, 817, 3862, then H already violates it.*

Proof. If H satisfies the target then, being finite, it satisfies 3862; by Theorem 4 every one-generated submagma $\langle x \rangle$ is trivial or the shift; a violation at (x, s) would live in the sub-extension over $\langle x \rangle$, which satisfies the four targets by Theorems 1 and 3. $\square$

So constant-coefficient abelian cohomology never refutes any of these four implications from any finite base. Non-vanishing H^2 over other bases is real: H^2_{1518} is nonzero for the right projection on two elements and for every 1518-base of sizes 4 and 5, and such extensions do refute two-variable laws ($1518 \Rightarrow 632$ and $1518 \Rightarrow 879$ at carrier 6, from the base $[[0, 1, 2], [0, 1, 2], [1, 0, 2]]$ with fibre $\mathbb{Z}/2$), but by Theorem 5 they cannot touch the one-variable targets. The remaining prose in this section is the finite-base equivalence of the four targets and the assembly of Theorem 5.

7 Verification status

Layer	Status
Certificate identities $C D_{1518} = D_{E'}$ in R	Lean 4.33.1 kernel, <code>decide</code> on integer coefficient lists, file <code>L2Cert.lean</code> ; axioms: <code>propext</code> only
Homotopy identity $EP + QD_{1518} = I_9$ in R (Theorem 3) $H^2 = 0$ numerically	Lean 4.33.1 kernel, file <code>H2Cert.lean</code> ; axioms: <code>propext</code> only; also re-multiplied in Python $\dim Z^2 = \dim B^2 = 2$ at all 39 prime specialisations; $\mathbb{Z}/4$, $\beta = 1$: 16 cocycles, 16 coboundaries
Same identities	re-multiplication in R in Python

Layer (continued)	Status
Rows (4)–(5) are the cocycle conditions; R is the finite-fibre ring	written argument above; not formalized. Independently re-derived by a second program with noncommutative coefficients (words in α, β), specialised to R only at the end: the identities (2) and all rows agree exactly
One-generated classification (Theorem 4)	core Lean 4.33.1, <code>lean/OneGenerated1518.lean</code> : <code>table</code> , <code>words_in.T</code> with no axioms; <code>one_or_three</code> with the three standard axioms; transcribed from Vampire’s equational proofs
Target equivalences under 1518, finite bases	Vampire 5.1.0: twelve theorems with left-quasigroup axioms; equationally $3862 \Rightarrow 47, 614, 817$; not in Lean
Unreachable set of Section 5	ETP <code>finite_graph.json</code> (sha256 <code>d609274e...</code>), row 1518 decoded; control: the 5 finite implications pass the shift and R checks
Structure of <code>Refutation939</code>	exhaustive search over labellings of its three classes by $\mathbb{Z}/5$; 20^3 labellings give base-dependent affine form, none constant (Theorem 3 forbids it)
Numeric cross-check	at every root r of $b^5 + b^3 - b^2 - 1$ in $\mathbb{F}_p$, $p \leq 47$ (39 cases), fibre $\mathbb{Z}/p$, $\beta = r$, $\alpha = r^4 - r$: $\dim Z_{1518}^2 = 2$ and contained in each target space, by an independent numeric implementation
Control	law 359, which 1518 implies universally, is contained with a certificate; law 3 (idempotence) is not satisfied by the linear magma over R , as expected
Method check	the same computation over the affine base $3x + y + 1 \bmod 6$ with fibre $\mathbb{Z}/2$ reproduces the 12-element $879 \not\equiv 4065$ model
Census for Corollary 2	Mace4 2009-11A; sizes 2–4 complete labelled enumeration (audited against plain enumeration at sizes 2, 3), sizes 5–7 complete up to isomorphism (audited at size 4): one-generated classes at size 3: one (the shift); at sizes 4–7: none, out of 54, 632, 41 319, 8 393 390 models. Independently: z3 4.16, reachability encoding, sat only at size 3; labelled counts 1, 6, 54, 1500 at sizes 2–5 agree between the tools; z3 also gives unsat at size 8 (532 s). All enumerated 1518-magmas of size ≤ 7 satisfy 47, 614, 817, 3862

Layer (continued)	Status
Census, Lean-checked	<code>census_lean/Census1518_n.lean</code> , $n = 2, \dots, 8$: bitvector encoding of the table, the law at every pair and the reachability chain from an arbitrary generator; proved by <code>bv_decide</code> (bundled CaDiCaL, LRAT proof checked by Lean’s verified checker run natively; recorded as the per-theorem axiom <code>_native.bv_decide.ax</code> , the <code>native_decide</code> trust model, outside the lab’s allowlist). Size 3 is proved to be exactly the two labelled copies of the shift. Each file also carries encoding controls proved by kernel <code>decide</code> with no axioms: a genuine 1518 table of that size satisfies every encoded law hypothesis, and at size 3 the shift with generator 0 satisfies the reachability chain with <code>full = true</code> . The controls exist because a first size-8 file was vacuous ($8\#3$ wraps to $0\#3$, so its range hypotheses were false); the corrected size-8 file omits ranges, and its main theorem carries the SAT axiom like the others. Compile times 0.5 s to 68 s for sizes 2 to 7, 7:24.45 for size 8. <code>CensusBridge.lean</code> restates size 2 for functions $\text{Fin } 2 \rightarrow \text{Fin } 2 \rightarrow \text{Fin } 2$ and proves it by kernel <code>decide</code> with no native evaluation

A Lean proof of Theorems 1 and 3 themselves would need the extension defined generically, the derivation of the rows from the law terms, and Step 1. None of that is done.

8 Scope and prior art

Not covered: bases with a one-generated submagma of 9 or more elements; coefficients $\alpha_{x,y}, \beta_{x,y}$ varying with the base pair, which Section 5 shows do reach the targets over this very base; iterated extensions; fibres on which β is not injective.

No statement of Theorem 1 was found in the pinned ETP blueprint, paper, commentary or Lean sources, in the Lean Zulip *Equational* stream, in the SAIR Foundation Zulip, or on arXiv (searches dated 4 September 2026). What exists is the empirical record above. Originality is therefore unresolved, not claimed.

9 Questions

1. Is the vanishing $H_{1518}^2(\mathbb{Z}/3 \text{ shift}, M) = 0$ for all finite 1518-fibres, or the resulting containment, already known or written down somewhere?
2. Has the conjecture of 485148288 been settled beyond a census? With it, Corollary 2 would say that constant-coefficient cohomology can never refute these implications from any finite base.
3. Over the shift the obstruction does *not* survive base-dependent coefficients: **Refutation939** is such an extension. Has the cohomology with base-dependent coefficients (abelian objects in the slice category, in Bolan’s phrasing) been computed for the shift, so that the 15-element models are the generic case of something?

10 Reproduction

Standard-library Python, Mace4 and Lean 4.33.1; inputs pinned by commit. Outputs as run are recorded alongside the scripts.

- `fetch_inputs.sh`: pinned ETP law list and refutation tables; Parsimagma’s uncovered-pairs file.
- `run_mace4.sh`, `monogenic_1518.py`: the census behind Corollary 2.
- `l2_symbolic.py`: the ring, the cocycle rows, the lattice containment, the numeric cross-check; run once for each base.
- `l2_symbolic.py`, `h2_vanishing.py`: the certificates for Theorems 1 and 3; `L2Cert.lean`, `H2Cert.lean`: their kernel checks.
- `rederive_rows.py`: independent noncommutative re-derivation of the rows.
- `z3_census.py`: the census with z3.
- `unreachable_set.py`: the 13 laws from the ETP finite graph.
- `witness939_varying.py`: the structure of `Refutation939`.
- `gen_census_lean.py`, `census_lean/`: the Lean census proofs and their README.